8.3 Finding the Missing Dimension of a Rectangular Prism

Lesson Introduction

Benchmark	MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.	Learning Target	• I can find the value of a missing dimension of a rectangular prism.

	Building On	Working Toward
Benchmark Coherence	MA.5.GR.3.3 Solve real-world problems involving the volume of right rectangular prisms, including problems with an unknown edge length, with whole-number edge lengths using a visual model or a formula. Write an equation with a variable for the unknown to represent the problem.	MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

Essential Questions	• What formulas or strategies can you use to find the missing dimension of a rectangular prism? • How does knowing the volume of an object help find its dimensions?
Lesson Narrative	In this lesson, students build on their prior knowledge of solving real-world problems involving volume to find a missing dimension in a rectangular prism. Students collaborate with a partner to generate ideas on strategies to solve for missing dimensions, then solidify understanding through guided instruction and practice with given formulas for finding the missing dimensions. Students reflect on their understanding of finding volume and missing dimensions in the 8.3.6 Wrap-Up.

Component Summary	8.3.1 Warm-Up (~5min.)	Critical-thinking activity to find various solutions to area problem *(SE p. 16)*	8.3.4 Guided Practice (10 min.)	Guided practice finding missing dimension in context *(SE p. 19)*
	8.3.2 Collaboration (5 min.)	Collaboration on solving a real-world volume problem with a missing dimension *(SE p. 17)*	8.3.5 Your Turn! (5 min.)	Independent practice finding missing dimension *(SE p. 19)*
	8.3.3 Guided Instruction (15 min.)	Students explore a strategy to find missing dimensions of a prism *(SE p. 17)*	8.3.6 Wrap-Up (~5 min.)	Self-reflection on solving problems involving volume *(SE p. 20)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Face Additional Terms: • Volume • Dimension	• Four-Function calculator • (Optional) Isometric dot paper • (Optional) Unit cubes or math manipulatives • (Optional) Graphic organizer	• (Optional) Create a graphic organizer as explained in the Differentiate Instruction for students to use as a scaffold for recognizing and using given dimensions to find the missing values.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may forget to find the product of the two dimensions given to help find the missing dimension.
Students may not recognize that they need to divide the volume by product of the given dimensions to find the missing dimension.
Students may struggle to differentiate between the measurement of a one-dimensional length, the area of a two-dimensional face, and the volume of a three-dimensional figure.</td><td>Consider using a table to help students keep track of length, width, and height for each volume problem. This may help reinforce the need for all three units in solving problems.
Provide multiple opportunities to highlight the use of units in area and volume to reinforce the distinction between the two measurements.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Compare and Connect
8.3.3 Guided Instruction positions students to notice similarities between the structures of two formulas to find the missing dimension. Position students to recognize how both formulas work using the inverse operation (division) to find the missing factor and to connect this formula to previous knowledge on finding volume.</td><td>MA.K12.MTR.6.1: Reasonableness
Students will be working to solve real-world problems where they must find the measurement of a missing dimension in context. Consider encouraging students to predict solutions before solving the problems and use estimation to confirm the reasonableness of their solutions.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">In 8.3.4 Guided Practice, consider providing isometric dot paper or math manipulatives for students to visualize or create the prisms in the real-world story problems to see the missing dimension.

Consider providing graphic organizers to keep track of the dimensions or formulas with blank lines for students to fill in given values as scaffolded steps for students to fill in to assist in solving. Directly modeling and using a think aloud on how to substitute given values into the volume formula can also support auditory approaches to reasoning.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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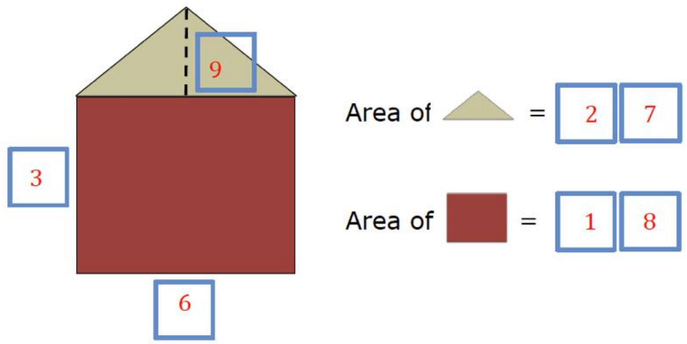
8.3.1 Warm-Up: Area of a Wall

Component Overview: *Prompt students to find dimensions of the rectangle and triangle and their corresponding areas using the digits zero through nine once each at most, clarifying that there are multiple solutions.*

8.3.1 Warm-Up: Area of a Wall

1. Using the digits $0 - 9$, at most one time each, fill in the boxes to make a true statement.
   Answers will vary.



Consider supporting students who have difficulty getting started by reviewing how to find the area of a rectangle or triangle. For the rectangle portion, prompt students to consider what digits one through nine are factors of products that are also made up of digits one through nine. For the triangle, prompt students to recognize the length of the rectangle is the same measurement as the base of the triangle.

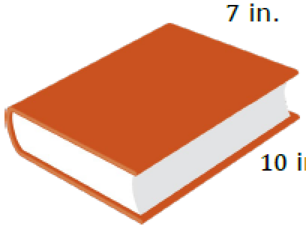
For an additional challenge, encourage students to find as many solutions as possible.

8.3.2 Collaboration: Missing Measurement

Component Overview: *Students collaborate on finding the missing measurement of a prism within the real-world context of a missing length of a textbook. Have students work in partners to calculate the volume of the old textbook by applying the volume formula, then dividing by the product of the given dimensions to find the missing dimension.*

8.3.2 Collaboration: Missing Measurement

Jolene works for a textbook company filling orders. She has discovered each textbook has a width of 7 inches (in.), a height of $2\frac{2}{3}$ in., and a length of 10 in.



The textbook company has decided to make the textbooks smaller by reducing the length. The new volume of the textbook will be 168 in^3.

1. Complete the table.

Old Textbook	New Textbook
$V = l \times w \times h$	$V = l \times w \times h$
$V = 10 \times 7 \times 2\frac{2}{3}$	$168\text{ in}^3 = l \times 7\text{ in.} \times 2\frac{2}{3}\text{ in.}$
$V = 186\frac{2}{3}\text{ in}^3$	$168\text{ in}^3 = 9\text{ in.} \times 18\frac{2}{3}\text{ in}^2$

2. Work with your partner to determine the new length of the textbook. Explain your process.
   Answers will vary. I made an equation where length was the variable and I multiplied width times height and then divided 168 by my answer to get my new length. I multiplied $7 \cdot 2\frac{2}{3}$ *which equals* $18\frac{2}{3}$*. Then I divided* $168 \div 18\frac{2}{3}$ *which equals* 9*.*

Scaffold student thinking as they complete the table in question 1 by asking:

- *What values represent length, width, and height in the old textbook? How do you know?*
- *Which value did you write in the New Textbook column as the inches-squared value? Why?*
- *How can you use the volume formula and the old textbook information to help you find the new length of the textbook?*

Compare and Connect
Question 2 provides an important opportunity for students to compare the volume formula and make connections to how to use the inverse operation (division) to find the missing value. Position students to connect the equation for the new textbook to the volume formula and recognize that they are looking for a missing factor. Consider having students share their process with another partnership or to the class to address any misconceptions.

For struggling students, connect how to use division to find missing factors with simpler problems, like how we might find the missing factor in the equation $5 \times 2 \times a = 20$.

8.3.3 Guided Instruction: Finding a Missing Dimension

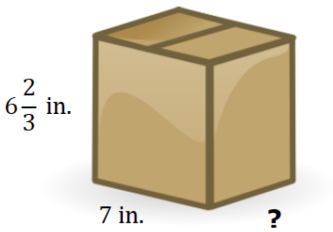
Component Overview: *Students are guided through two formulas or methods for finding a missing dimension on a rectangular prism when given the volume.*

8.3.3 Guided Instruction: Finding a Missing Dimension

Each flat surface of a prism is called a face.

When only two dimensions of a three-dimensional prism are given, the product of the two given dimensions represents the area of that face of the prism.

1. What is the area of the front face of this prism? Include units.



$46\frac{2}{3} in^2$

The volume of the prism is equal to the area of the face multiplied by the length of the missing dimension. The value of the missing dimension can then be calculated by dividing the volume by the area of the face.

$$V = Area\ of\ the\ face\ \times measurement\ of\ the\ missing\ dimension$$

$$Measurement\ of\ the\ missing\ dimension = \frac{V}{Area\ of\ the\ face}$$

For question 1, encourage math connections between vocabulary and volume concepts by asking:

- *What shapes are the faces of a rectangular prism?*
- *How do you find the area of a face of a rectangular prism?*
- *How can $V = lwh$ and $V = Bh$ both be used to find the volume of a rectangular prism and a missing dimension?*

Compare and Connect
Position students to compare and connect the two formulas for finding a missing dimension. Consider having students Think-Pair-Share with prompts like:

- *What is the same or different about these formulas?*
- *Can you use either formula to solve the problem?*
- *How are these formulas related to the formulas used to find volume: $V = lwh$ and $V = Bh$?*

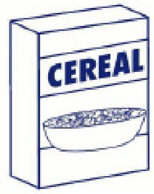
8.3.3 Guided Instruction: Finding a Missing Dimension (continued)

2. If the volume of the box is $233\frac{1}{3}$ in^3, what is the value of the missing dimension of the box?

$V_{\text{box}} = \frac{700}{3}\,in^3 = 7\text{ in} \times w \times \frac{20}{3}\text{ in}$

$w = \dfrac{\frac{700}{3}\,in^3}{\frac{140}{3}\,in^2} = 5\text{ in}$

3. Rafael was trying to find the missing dimension of the cereal box shown.



30 cm

20.3 cm

Volume = 1,218 cm^3

He made a mistake while solving for the width of the box, but he cannot find it. Inspect his work below.

Rafael's Work
20.3 cm $\times\ 30$ cm $\overline{60.9\ \text{cm}^2}$ Area of face $20\ \text{cm}^3$ Volume $60.9\overline{)12180}$ $\underline{1218}$ 0

Write a note to Rafael on how to correct his error(s).

Raphael, it looks like you forgot to multiply by zero first when multiplying 20.3 times 30. This gave you the wrong area of the face, so your division was also incorrect. In addition, the units for your answer should be cm because you're only finding the width and dividing cm^3 by cm^2.

Consider modeling how to substitute values from question 1 into the volume formula when finding the missing dimension in question 2. It may be helpful to first substitute into the volume formula, showing the unknown variable as the measurement of the missing dimension, and then into the equation from question 2.

Prompt students to discuss the reasonableness of Rafael's answer while fostering a growth mindset by asking:

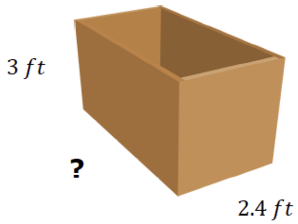
- *Notice Rafael's solution is the volume is 20 cm^3. Was he trying to find the volume or something else? Does the value he found make sense for the missing dimension?*
- *Look closely at Rafael's multiplication. What do you notice about the product and the factors? Is 60.9 cm^2 a reasonable answer to 20.3×30? Why or why not?*
- *How could you give kind, helpful, and specific feedback to Rafael to check his arithmetic? How could you help him consider a growth mindset when making mistakes?*

8.3.4 Guided Practice: Find the Missing Dimension

Component Overview: *Students practice finding the missing dimensions of rectangular prisms within two real-world contexts. Consider working through these problems as a class to address any remaining misconceptions.*

8.3.4 Guided Practice: Find the Missing Dimension

1. Sally broke her ruler before she finished measuring her toy chest. She knows the toy chest has a volume of $25.2\ ft^3$.



What is the length of the toy chest?

$l = 3.5$ ft

2. Damion works in construction and needs to cut a wooden beam to fill a cement hole. He knows that the beam has a width of 1.2 meters (m) and a length of 0.25 m. If Damion knows that the volume of the beam needs to be 1.5 cubic meters (m^3), what is the height?

$h = 5$ m

Encourage students to make predictions for measurements of the missing dimensions. Consider modeling through a think-aloud how to make predictions. For example, "The toy chest has a volume of $25.2\ ft^3$. Multiplying the width and height gives an area of $7.2\ ft^2$. The missing dimension should be around $3ft$ since $7 \times 3 = 21$."

Provide a labeled image or create a rough drawing of a wooden beam to help students recognize the shape as a rectangular prism. Then scaffold student thinking for question 2 when solving as a class by asking:

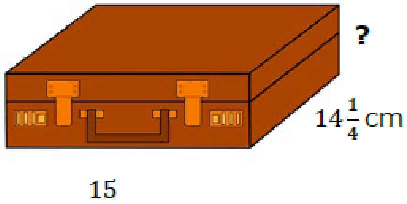
- *What formula should we use to find the missing dimension? Why?*
- *How do we calculate the area of the face?*
- *If we know the area of the face and volume, what operation is needed to find the measurement of the missing dimension?*
- *Is our solution reasonable? How can we use estimation to find out?*

8.3.5 Your Turn!

Component Overview: *Students independently practice finding missing dimensions given two dimensions and the volume.*

8.3.5 Your Turn!

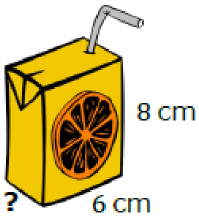
1. Mr. Rodrigo knows that his briefcase has a volume of $1{,}068\frac{3}{4}$ cubic centimeters (cm³). He labeled the dimensions of the briefcase that he knew.



What is the height of the briefcase?

$h = 5$ cm

2. Victor works in a factory that makes juice boxes. He knows that the juice boxes have a height of 8 centimeters (cm), a length of 6 cm, and a volume of 192 cubic centimeters (cm³).



What is the width of the juice box?

$w = 4$ cm

Consider providing a graphic organizer that prompts students to write the given values into the volume formulas for finding a missing dimension from 8.3.4 Guided Practice question 2. This can help students recognize what values are missing and how to solve. The organizer could be laminated, and students could use a dry-erase marker to write, erase, and rewrite given dimensions and values in the formula for all problems.

Consider providing additional challenge problems for students who may finish early or are ready for higher cognitive demand.

- *How could you change the dimensions of the juice box so that it could hold double the amount of juice? Half?*
- *A swimming pool can hold $2250 ft^3$ of water. The length and width are in a ratio of $1:2$. The height and length are in a ratio of $1:3$. Find the dimensions of the swimming pool.*

8.3.6 Wrap-Up: Reflection

Component Overview: *Students reflect on and rate their understanding from this and previous lessons on how to find volume of a rectangular prism and how to find a missing dimension of a rectangular prism. Consider alternative methods for students to share their thinking and explanations for the ratings.*

8.3.6 Wrap-Up: Reflection

1. Rate your strength on the following areas using the scale.
   Answers will vary.

   1 – still learning
   2 – making progress
   3 – got it

Skill	Rating
Finding the volume of a rectangular prism.	3
Find a missing dimension of a rectangular prism.	2

2. Write an explanation of why you rated yourself the way you did.
   Answers will vary. I can find the volume by multiplying length times width times height. Sometimes I get confused solving for a missing dimension when fractions are involved.

For students who might struggle representing their thoughts through writing, consider alternative ways for students to provide their reflection reasoning. Students could choose to record themselves explaining their reasoning verbally, drawing and labeling, or providing an example.